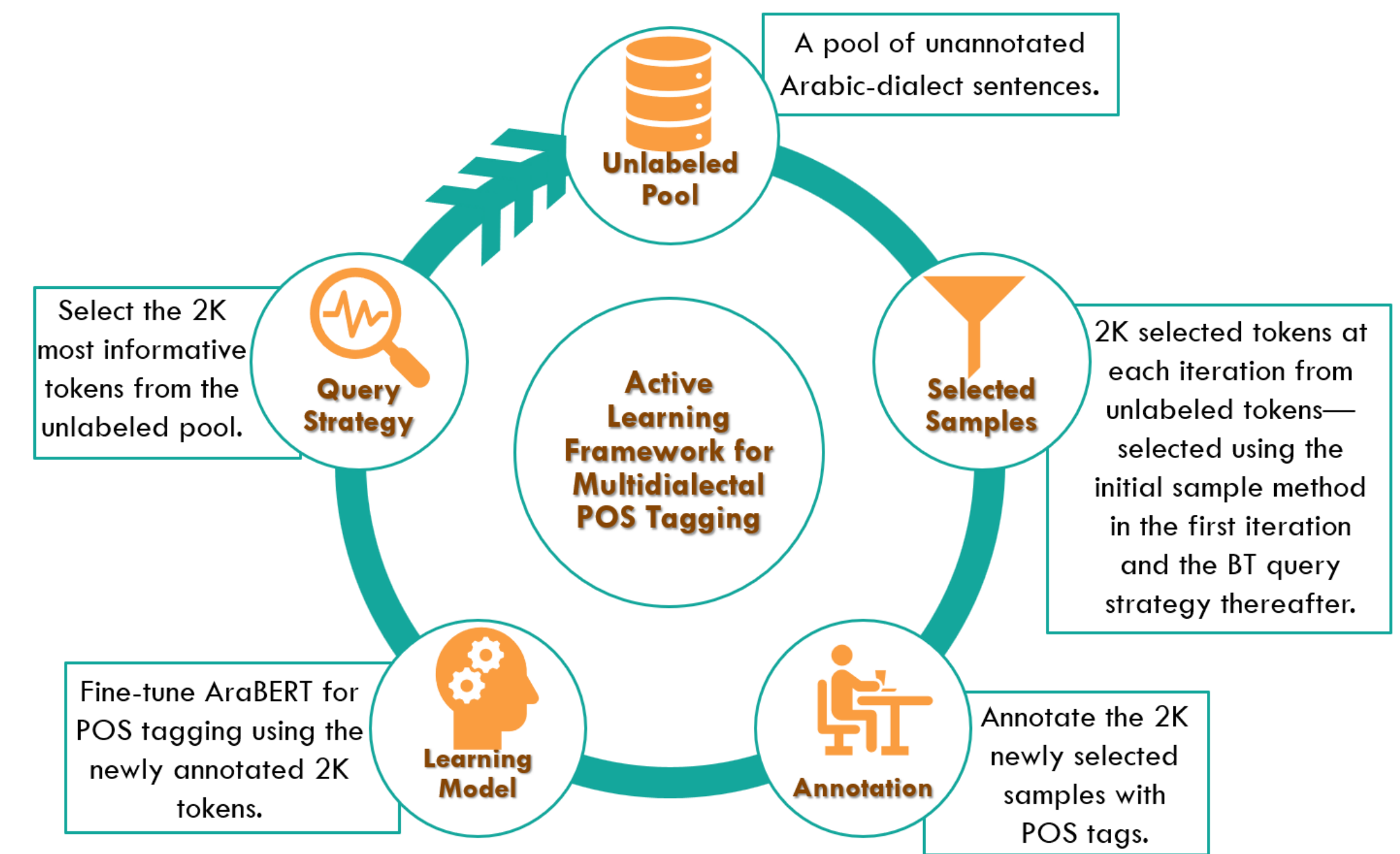


Contributions

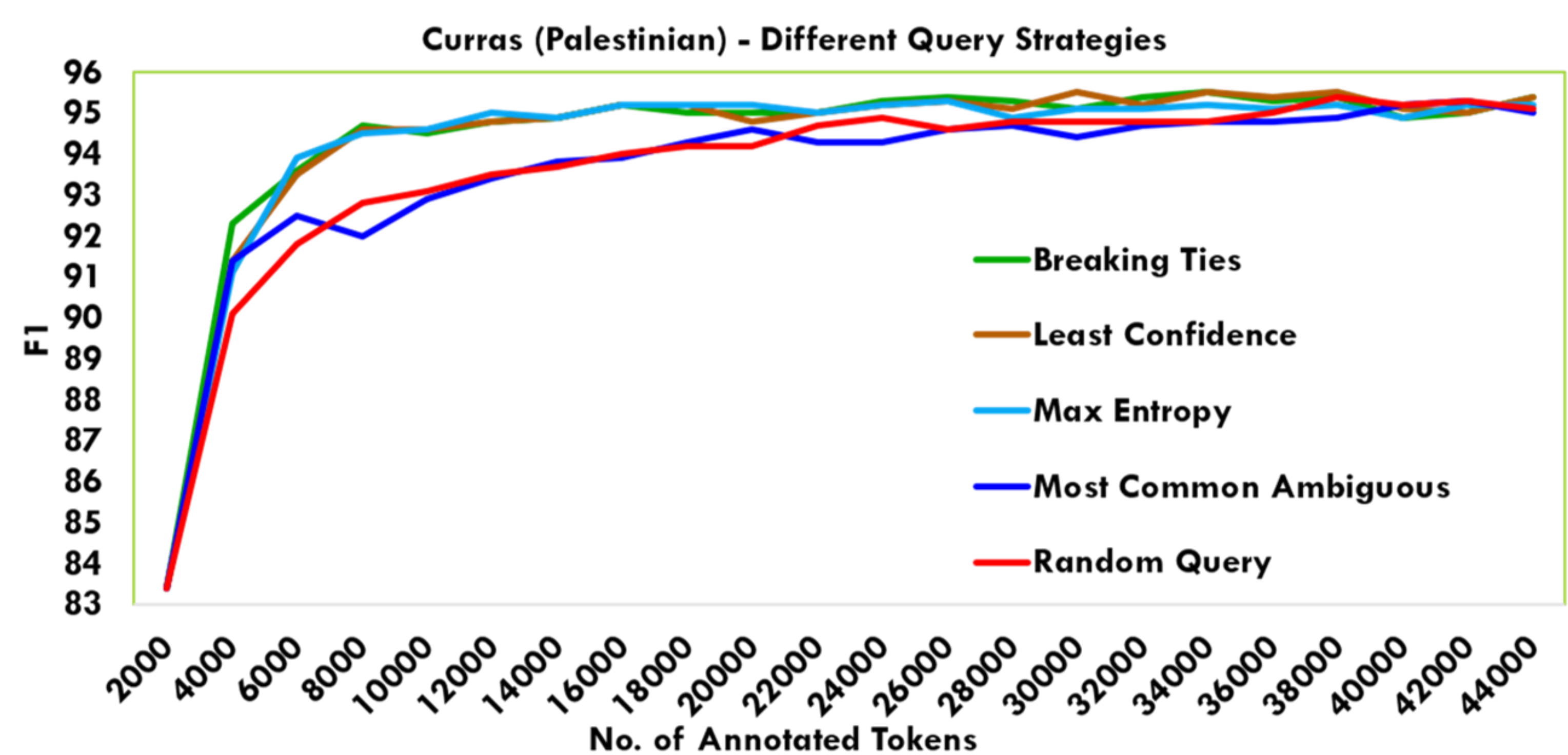
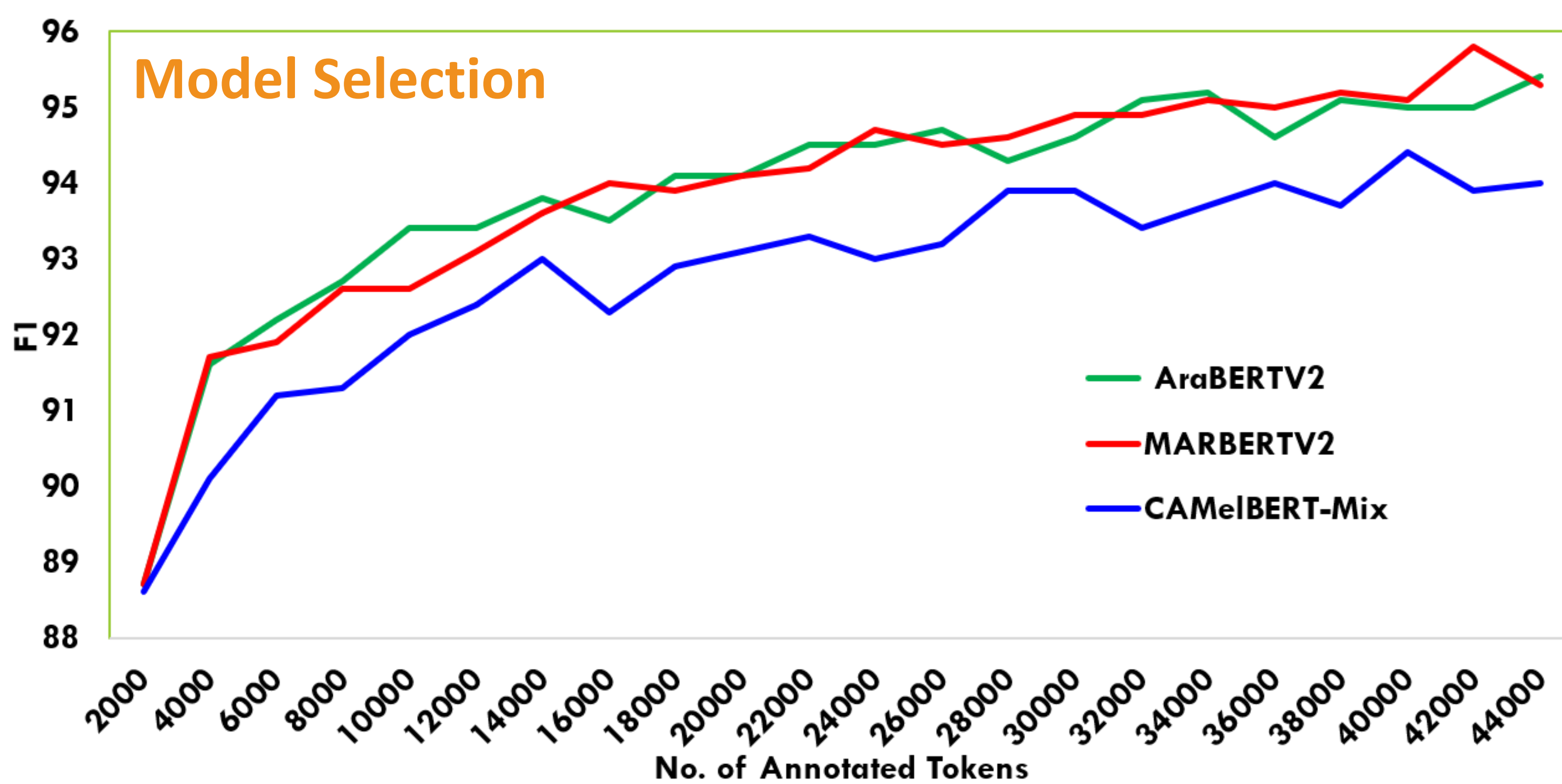
- Evaluation of active learning for multidialectal (Palestinian, Syrian, Lebanese, Egyptian) Arabic POS tagging (29 experiments)
- Insightful analysis and 10 assessments for initial sample selection across dialects.
- Active learning pipeline, e.g., achieved 97.6% with only 16,000 tokens in Emirati.



Identifying Best Active Learning Strategy

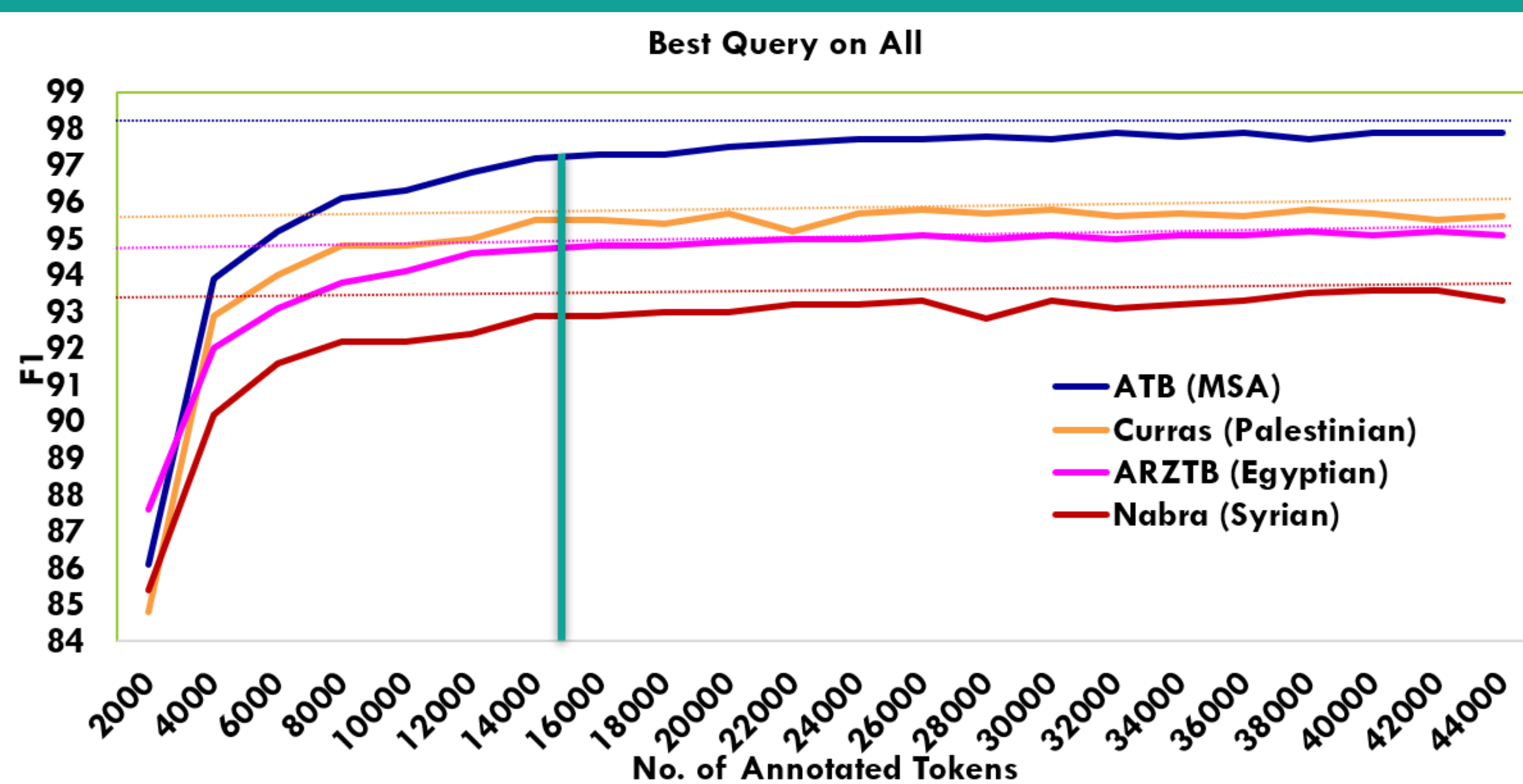
Methodology for identifying the most effective query strategy:

1. Evaluate various pretrained learning models
2. Evaluate different initial sample selection methods
3. Evaluate different annotation level strategies
4. Evaluate different query strategies



- Best learning model: AraBERTV2
- Best Initial Sample Selection : Most-Dissimilar-Sentences
- Best Annotation Level Strategy: Mask-All-Unknowns
- Best Query Strategy: Breaking Ties

Conclusions



- **15,000 tokens** are enough to build **POS tagger** since **performance stabilizes** at that point.
- **Why:** embeddings distribution of **15K** (from Curras) shows coverage of **all POS tags** - the model has learned word patterns across all POS tags

- **Colored points** are clusters of all words (56K) in the corpus. **Black points** are those that are selected (15K) during learning.
- More **ambiguous tokens** are more likely to be **selected** in training (**dense area**)



Paper